

Building a Mini GPT from Scratch: Findings

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1. Setup and Data

The corpus is *Shakespeare.txt* (1,115,394 characters). Character-level tokenization gives a vocabulary of **65** symbols (letters, digits, punctuation, space and newline), stored in `char_to_idx / idx_to_char`. The first 90% of the text is used for training and the last 10% for validation. Training examples are sliding windows of length 128: the input is tokens `[t ... t+127]` and the target is tokens `[t+1 ... t+128]`, so the model predicts the next character at every position. With stride 32 this gives 31,367 training windows (each character appears in 4 windows at different positions). SEED = 1605 (SID4) is used for model initialization, data shuffling and all sampling.

2. Model Architecture

The model is a decoder-only GPT written from scratch. **Multi-head masked self-attention** is implemented by hand: a single linear layer produces Q, K and V, which are split into 8 heads of dimension 32. Scores $QK^T/\sqrt{d_k}$ are masked with a lower-triangular causal mask (future positions set to -inf) before the softmax. A unit test in the notebook confirms that changing future tokens leaves earlier outputs unchanged. Each **decoder block** computes $x = \text{LN}(x + \text{MHA}(x))$, then $x = \text{LN}(x + \text{FFN}(x))$, where $\text{FFN} = \text{Linear}(256 \text{ to } 1024)$, GELU , $\text{Linear}(1024 \text{ to } 256)$. The full model is: token embedding + learnable positional embedding, then 4 decoder blocks, then a linear projection to the 65 vocabulary logits. No `nn.Transformer` or `nn.MultiheadAttention` is used.

Setting	Value
Total parameters	3,225,153
Hidden dimension (<code>d_model</code>)	256
Attention heads	8 (head dim 32)
Decoder layers	4
Context length	128
Vocabulary size	65

3. Training

Training used **Adam** (learning rate $3e-4$), **batch size 64** (8,192 tokens per step), dropout 0.1, gradient clipping at 1.0, and **8 epochs** (490 steps per epoch, 3,920 steps in total) on an Apple-silicon GPU (about 23 minutes). The loss starts at 4.24, close to $\ln(65) = 4.17$ for a uniform guess, and falls to **1.40** (train) / **1.54** (validation), from 2.36 / 2.07 after epoch 1. Both curves are still decreasing at epoch 8 and the gap between them is small, so the model is not overfitting. More epochs would improve it a little further.

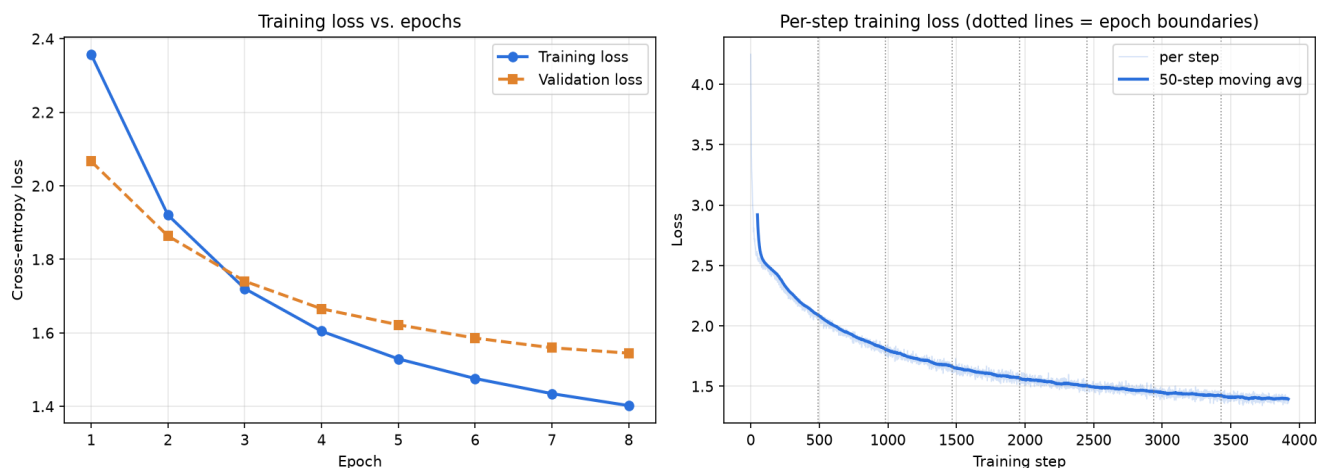


Figure 1. Left: mean training loss per epoch with validation loss. Right: per-step training loss.

Epoch	1	2	3	4	5	6	7	8
Train loss	2.358	1.920	1.720	1.604	1.529	1.476	1.435	1.402
Val loss	2.067	1.864	1.741	1.665	1.622	1.586	1.559	1.545

4. Sampling Results (prompt: "ROMEO:")

All samples use the prompt "ROMEO:" and generate 400 characters. Greedy decoding is deterministic; all sampled outputs use SEED = 1605. Each output below is truncated to fit the page (full text in the notebook). With the same seed, the $T = 1.0$ and $k = 30$ samples are identical, because every character $T = 1.0$ sampled was already in the top 30.

Greedy

ROMEO:

The present of the princess of the princess.

POMPEY:

I will not so much a strike of the princely of the country.

KING RICHARD III:

The princely of the princess of the prince,
And the present of the present of the prince,
And the present of the princess of the world.

KING RICHARD III:

The present of the princess of the ...

Temperature T=0.3

ROMEO:
The king and the present of the body of the fearful of the king.

MENENIUS:
The law may we have been the princely words,
That we we may be the world and all the head.

CORIOLANUS:
I will not be the father's comes of the court?

CORIOLANUS:
He is the proclaim to the world of me?

CORIOLANUS:
Be so, sir, I have been your an ...

Temperature T=1.0

ROMEO:

DUKE VINCENTIO:
You have life, as I may befaect;
And go thy ghost looks power than may in hum.
If With come men my of with comforts,
And with we lady in his spare offence.

SEMN:
It is doubt:
Hark! yet cousin.

WARWICK:
And, but now, I shall be spit and son to our
In encling more marry subjects, than what?

ROMEO:
More, ...

Temperature T=1.8

ROMEO: traF Bylorow'd speit his; like,
ash; nan hifae, nor hag?
Gon anot-glaced. Gauldera, mad? wert is my reR:
My hang my of ANNEjIol:
Resirous your well.
Now, 'PovPETHMA, he Dpinor to the mark's tragto.
Ho.
Faply:' you
I was life, Tuke moorsal exprept for fools.
Tenco?

ISABELLY:
Will
I swear in Badnist: if York'?

NORFOLK:
Hi ...

Top-k k=3

ROMEO:
Traitor, to-dear to the counterfacte of thine
That that we have sent my brother's head;
There is not the poison, and took me whose:
Though thou shalt by the proceed of their court
And so such a provide and with a servant:
If, the should she was the world of the prince,
To brother hearts the was to said of him.

KING RICHA ...

Top-k k=30

ROMEO:

DUKE VINCENTIO:
You have life, as I may befaect;
And go thy ghost looks power than may in hum.
If with come men my of with comforts,
And with we lady in his spare offence.

SEMN:
It is doubt:
Hark! yet cousin.

WARWICK:
And, but now, I shall be spit and son to our
In encling more marry subjects, than what?

ROMEO:
More, ...

Quantitative comparison

To support the qualitative comparison, each decoder generated 1,000 characters (averaged over 3 seeds for the stochastic methods). **Distinct-2** = unique word bigrams / total bigrams (higher means more diverse).

Repeated 4-grams = share of word 4-grams that occur more than once (higher means more looping).

Real-word rate = share of generated words that appear in the training text (a coherence proxy). **Model NLL** = per-character negative log-likelihood of the generated text under the trained model (lower means more predictable).

Decoder	Distinct-2 (diversity)	Repeated 4-grams	Real-word rate	Model NLL
Greedy	0.154	0.939	1.000	0.660
Temperature T=0.3	0.744	0.065	0.995	0.829
Temperature T=1.0	0.993	0.000	0.878	1.330
Temperature T=1.8	1.000	0.000	0.477	2.608
Top-k k=3	0.936	0.000	0.958	1.018
Top-k k=30	0.986	0.000	0.911	1.295

5. Analysis: Comparing Decoding Techniques

Greedy decoding always takes the most likely next character. It is deterministic and gives the lowest model NLL (0.66). Every word is a real word and speaker turns are formatted correctly (POMPEY:, KING RICHARD III:), but it quickly falls into a **repetition loop**, cycling through "the present of the princess of the prince..." again and again. That gives the lowest Distinct-2 (0.15) and 94% repeated 4-grams. Because greedy search

optimizes each step locally, it cannot leave a high-probability cycle once it enters one.

Temperature sampling samples from $\text{softmax}(\text{logits} / T)$. At $T = 0.3$ the distribution is sharpened: the output is conservative and almost perfectly spelled (99.5% real words), with plausible speaker turns (MENENIUS:, CORIOLANUS:, CAPULET:). It escapes greedy's loop but still reuses safe phrases (6.5% repeated 4-grams). At $T = 1.0$ we sample the model's own distribution: varied play-like dialogue with some invented words ("bafaect", "encling", "unhoods"). Distinct-2 is 0.99 and 88% of words are real. At $T = 1.8$ the distribution is flattened: random capitals and rare characters appear ("traF", "ANNEjloI", "PovPETHMA"), only 48% of words are real, and the NLL jumps to 2.61.

Top-k sampling samples only from the k most likely characters. With $k = 3$ the text is coherent and non-repetitive (96% real words, no repeated 4-grams, Distinct-2 = 0.94): "Traitor, to-dear to the counterfacte of thine / That that we have sent my brother's head". With $k = 30$ the output is almost identical to $T = 1.0$; the 400-character sample is exactly the same, because the top 30 of 65 characters already hold nearly all of the probability mass. Over longer 1,000-character runs the cutoff helps slightly (91% vs. 88% real words).

Most coherent: low-temperature sampling ($T = 0.3$), with top-k $k = 3$ a close second. $T = 0.3$ has the highest real-word rate (99.5%) and correct play structure, and it avoids greedy's loop. Top-k with $k = 3$ is slightly less accurate (96%) but has no repetition, so it is the best choice for text that is both coherent and varied. Greedy has the highest per-step likelihood and perfect spelling, but its output as a whole is less coherent because it repeats one phrase over and over.

Most diverse: high-temperature sampling ($T = 1.8$), which has the highest Distinct-2, no repeated n-grams, and the most unusual characters and words. The price is coherence: fewer than half of its words are real. Among readable settings, $T = 1.0$ and top-k with $k = 30$ are the most diverse.

Conclusion. Decoding trades coherence against diversity. Greedy decoding and low T / small k concentrate probability on likely tokens, which produces coherent but repetitive text. High T / large k spread probability out, which produces diverse text that drifts into nonsense. Top-k makes this trade-off controllable by cutting off the unreliable tail of the distribution, but with a small character vocabulary k has to be small to change the behavior much.